



Question Bank with Answers

Chapter 3 — Web Applications

Exam-style practice questions

PROGRAMMING AND ARTIFICIAL INTELLIGENCE | SEMESTER 1

115 Questions Across 4 Sections

Fill in the blank • True/False • Matching • Multiple choice • Short answer • Analysis

QUESTION BANK

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Every question is reproduced exactly as written, grouped by lesson. Each question is tagged by format (fill in the blank, true/false, matching, multiple choice, short answer, analysis, or extended response) with its answer shown directly beneath in a green box.

3-1 • CHAPTER 3 QUESTION BANK

3-1

The Overall Structure of Web Applications

Key idea: a web application uses three cooperating tiers — frontend, backend, and database.

FILL IN THE BLANK

Q1

A web application is broadly composed of three tiers. The screen component that users see directly is called the (a). The part responsible for server-side processing that users do not see is called the (b). A system that organizes, stores, and manages data is called the (c).

ANSWER

- (a) Frontend
- (b) Backend
- (c) Database

TRUE / FALSE

Q2A

The frontend is the screen component that users see and interact with directly. (True or False?)

ANSWER

True

TRUE / FALSE

Q2B

The backend performs server-side processing that users do not see. (True or False?)

ANSWER

True

TRUE / FALSE

Q2C

The database is mainly responsible for choosing the colors and layout of a web page. (True or False?)

ANSWER

False — the frontend/CSS handles appearance; the database stores and manages data.

TRUE / FALSE

Q2D

The three tiers cooperate to make a web application work. (True or False?)

ANSWER

True

MATCHING

Q3

For each of a–f, indicate which tier it belongs to: A Frontend, B Backend, C Database.

- a. Product listing page
- b. Stock checking processing
- c. Storage of customer information
- d. Cart screen
- e. Payment processing
- f. Storage of order history

ANSWER

- a → A (Frontend)
- b → B (Backend)
- c → C (Database)
- d → A (Frontend)
- e → B (Backend)
- f → C (Database)

MULTIPLE CHOICE

Q4

Choose the one that most appropriately describes the frontend.

- A. A system that stores customer information
- B. The screen component users see and interact with directly
- C. A server-side processing system
- D. A communication protocol

ANSWER

B

FILL IN THE BLANK

Q5

Complete the application flow. A user enters a product search keyword. The (a) receives the keyword and sends it to the (b). The backend searches the (c). The database returns matching data. The backend organizes the result and sends it to the frontend, which (d) the result.

ANSWER

- (a) frontend
- (b) backend
- (c) database
- (d) displays/shows

SHORT ANSWER

Q6.1

Why is the frontend important?

ANSWER

It is the part users see and interact with directly, and it collects user input.

SHORT ANSWER

Q6.2

What is the role of the backend?

ANSWER

It performs server-side processing, processes data, and makes decisions.

SHORT ANSWER

Q6.3

What is the role of the database?

ANSWER

It organizes, stores, manages, retrieves, and records data.

SHORT ANSWER

Q6.4

Why must the three tiers cooperate?

ANSWER

Each tier has a clear responsibility, and together they complete the user's request and result.

6-MARK

Q7 6 marks

Explain how the three tiers of a web application cooperate when a user posts a photo on a social media app. Refer to: the role of the frontend, backend, and database.

ANSWER

The frontend provides the photo-upload screen and collects the user's action. The backend receives the request and performs server-side processing. The database stores and manages information related to the post and user. The three tiers cooperate so the photo can be submitted, processed, stored, and later displayed.

THINK AS AN ENGINEER

Q8.1

A clinic wants a web app where patients book appointments, check which time slots are free, and see their past visits. Assign the work: which tier handles the booking form, checking free time slots, and storing past visits?

ANSWER

Booking form → frontend. Checking available slots → backend using data from the database. Storing past visits → database.

THINK AS AN ENGINEER

Q8.2

The booking page opens normally, but no free slots appear. Which tier is clearly working, and which one has probably failed?

ANSWER

The frontend is clearly working because the page opens. The backend or database may have failed because the slot information is not being processed or retrieved.

THINK AS AN ENGINEER

Q8.3

Decide: should the clinic separate the app into three tiers? Give two reasons.

ANSWER

Yes. Separation gives each tier a clear responsibility and makes development, maintenance, and troubleshooting easier.

NEW CONTEXT

Q9

A weather app lets users type a city and see the forecast. Using the three-tier model, name the tier that takes the city name, looks up the forecast data, and stores past forecasts. Identify one job the frontend cannot do on its own.

ANSWER

Taking the city name → frontend. Looking up/processing forecast data → backend. Storing past forecasts → database. The frontend cannot perform the server-side processing or manage the stored database data on its own.

REFLECT & CHALLENGE

Q10.1

Reflect: Why would putting frontend, backend, and database responsibilities into one undivided part make a large web application harder to manage?

ANSWER

Different responsibilities would be mixed together, making development, maintenance, and troubleshooting harder.

REFLECT & CHALLENGE

Q10.2

Challenge: Name one web application you use and describe one task done by each tier.

ANSWER

Example: an online shopping site — frontend shows products and collects searches; backend processes searches/orders; database stores product, customer, and order information.

3-2 • CHAPTER 3 QUESTION BANK

3-2

Web Application Communication Methods

Key idea: client requests, server responds — over HTTP/HTTPS, with methods, status codes, APIs, and often JSON.

FILL IN THE BLANK

Q1

A web application operates by communication between a (a), which requests data or processing from the server, and a (b), which receives the request, processes it, and returns a response. This communication uses a protocol called (c). The protocol that adds SSL/TLS encryption to (c) is called (d).

ANSWER

- (a) Client
- (b) Server
- (c) HTTP
- (d) HTTPS

TRUE / FALSE

Q2A

A browser can act as a client. (True or False?)

ANSWER

True

TRUE / FALSE

Q2B

A server receives requests, processes them, and returns responses. (True or False?)

ANSWER

True

TRUE / FALSE

Q2C

HTTPS removes encryption from HTTP. (True or False?)

ANSWER

False — HTTPS adds SSL/TLS encryption to HTTP.

TRUE / FALSE

Q2D

GET retrieves data, while POST sends data to a server. (True or False?)

ANSWER

True

MATCHING

Q3

For each operation (a–d), indicate which HTTP method is used: A GET, B POST.

- a. Displaying a web page
- b. Entering an ID and password into a login form and submitting it
- c. Displaying search results
- d. Submitting the contents of a contact form

ANSWER

- a → A (GET)
- b → B (POST)
- c → A (GET)
- d → B (POST)

MULTIPLE CHOICE

Q4

From the following options, choose the one that most appropriately describes HTTPS.

- A. A method for retrieving data
- B. A system for exchanging data between software
- C. A communication protocol that adds SSL/TLS encryption to HTTP
- D. A status code indicating a server problem

ANSWER

C

SHORT ANSWER

Q5.1

What does 200 indicate?

ANSWER

The request was processed successfully.

SHORT ANSWER

Q5.2

What does 404 indicate?

ANSWER

The requested page was not found / the URL does not exist.

SHORT ANSWER

Q5.3

What does 500 indicate?

ANSWER

A server-side error occurred.

SHORT ANSWER

Q6.1

What is an API?

ANSWER

A system for exchanging data between different pieces of software.

SHORT ANSWER

Q6.2

What does JSON stand for?

ANSWER

JavaScript Object Notation.

SHORT ANSWER

Q6.3

Why is JSON useful in web applications?

ANSWER

It is a data format that is easy for humans to read and computers to process.

MULTIPLE CHOICE

Q7.1

Which HTTP method retrieves a forecast?

- A. GET
- B. POST
- C. 404
- D. HTTPS

ANSWER

A

MULTIPLE CHOICE

Q7.2

Which method is appropriate for submitting a contact form?

- A. GET
- B. POST
- C. 404
- D. 500

ANSWER

B

MULTIPLE CHOICE

Q7.3

Which status code means the requested page was not found?

- A. 200
- B. 404
- C. 500
- D. 100

ANSWER

B

MULTIPLE CHOICE

Q7.4

Which status code indicates a server-side error?

- A. 200
- B. 201
- C. 404
- D. 500

ANSWER

D

MULTIPLE CHOICE

Q7.5

Which statement best describes an API?

- A. A screen design method
- B. A system for exchanging data between different pieces of software
- C. A database table
- D. A status code

ANSWER

B

6-MARK

Q8 6 marks

Explain how a browser and a server communicate when a user opens a web page. Refer to client, server, protocol, request, and response.

ANSWER

The browser acts as the client and sends a request to the server. The communication uses HTTP or HTTPS. The server receives and processes the request and returns a response. The browser receives the response and displays the requested page.

THINK AS AN ENGINEER

Q9.1

A student is building a login page and wants passwords protected while being sent to the server. Should the page use HTTP or HTTPS? Give a reason.

ANSWER

HTTPS, because it adds SSL/TLS encryption and helps protect communication content.

THINK AS AN ENGINEER

Q9.2

Which HTTP method is appropriate for sending the login data?

ANSWER

POST.

THINK AS AN ENGINEER

Q9.3

The page shows 'Not Found.' Which status code would help explain the problem?

ANSWER

404.

NEW CONTEXT

Q10

A weather service offers an API that returns today's forecast when a client sends a request. Name the HTTP method used to retrieve the forecast, why HTTPS should be used, and one advantage of JSON.

ANSWER

GET retrieves the forecast. HTTPS adds SSL/TLS encryption. JSON is easy for humans to read and computers to process.

REFLECT & CHALLENGE

Q11.1

Reflect: When a page fails to load, which status code would help you understand why, and what would it tell you?

ANSWER

404 indicates a missing page/URL; 500 indicates a server-side error. The correct code depends on the problem.

REFLECT & CHALLENGE

Q11.2

Challenge: Name one app you use and describe one request it sends and the response it expects.

ANSWER

Example: a weather app sends a request for a city's forecast and expects forecast data in the response.

3-3 • CHAPTER 3 QUESTION BANK

3-3

Fundamentals of Frontend Technology

Key idea: HTML structures, CSS styles, JavaScript behaves; semantic HTML, responsive design, and frameworks strengthen the frontend.

FILL IN THE BLANK

Q1

Web pages are mainly composed of three technologies: (a), which defines the structure of the page; (b), which specifies the appearance; and (c), which adds behavior and interactivity. The concept of adjusting the layout according to different screen sizes such as PCs, smartphones, and tablets is called (d).

ANSWER

- (a) HTML
- (b) CSS
- (c) JavaScript
- (d) Responsive design

TRUE / FALSE

Q2A

HTML defines the structure of a web page. (True or False?)

ANSWER

True

TRUE / FALSE

Q2B

CSS adds behavior to a web page. (True or False?)

ANSWER

False — JavaScript adds behavior and interactivity.

TRUE / FALSE

Q2C

Responsive design adjusts the layout according to different screen sizes. (True or False?)

ANSWER

True

TRUE / FALSE

Q2D

A framework can make development more efficient by providing commonly used functions and components. (True or False?)

ANSWER

True

MATCHING

Q3

Match each role (a–c) with A HTML, B CSS, C JavaScript.

- a. Defines structure
- b. Specifies appearance
- c. Adds behavior and interactivity

ANSWER

- a → A (HTML)
- b → B (CSS)
- c → C (JavaScript)

MATCHING

Q4

For each task (a–f), indicate A HTML, B CSS, or C JavaScript.

- a. Headings and paragraphs
- b. Changing text color and size
- c. Updating part of the screen when a button is pressed
- d. Placing images and links
- e. Setting the background color
- f. Validating form contents

ANSWER

- a → A (HTML)
- b → B (CSS)
- c → C (JavaScript)
- d → A (HTML)
- e → B (CSS)
- f → C (JavaScript)

SHORT ANSWER

Q5.1

What is semantic HTML?

ANSWER

Using element names/tags that indicate the meaning of each part of a web page, such as <header>, <nav>, <main>, and <footer>.

SHORT ANSWER

Q5.2

Name two benefits of semantic HTML.

ANSWER

Improved accessibility for screen readers and improved SEO because search engines can better understand page content.

MULTIPLE CHOICE

Q6

Choose the one that is NOT an appropriate benefit of semantic HTML.

- A. Screen readers can better understand the structure.
- B. Search engines can more easily understand content.
- C. The display speed necessarily becomes faster.
- D. Information becomes easier to convey to users with visual impairments.

ANSWER

C

SHORT ANSWER

Q7.1

What is responsive design?

ANSWER

A design concept in which the layout is automatically adjusted according to different screen sizes.

SHORT ANSWER

Q7.2

Why is responsive design needed?

ANSWER

Websites are accessed from different devices such as PCs, smartphones, and tablets.

SHORT ANSWER

Q7.3

What is mobile-first?

ANSWER

Prioritizing small screens such as smartphones from the design stage.

MULTIPLE CHOICE

Q7.4

Choose the reason responsive design is needed.

- A. Improve display speed
- B. Support various devices
- C. Raise search rankings
- D. Add behavior

ANSWER

B

SHORT ANSWER

Q8.1

What is a framework?

ANSWER

A structure that provides commonly used functions and components in advance to make web application development more efficient.

MULTIPLE CHOICE

Q8.2

Choose the best description of a framework.

- A. Language for page structure
- B. Structure for efficient web application development
- C. Encryption technology
- D. Screen-size design concept

ANSWER

B

REFERENCE

Q9

Major Frontend Frameworks

React	Developed by Meta; can efficiently update only part of the screen and is suited to large-scale web applications.
Vue	Simple structure; allows applications to be built incrementally from small features.
Next.js	Adds mechanisms to speed up page display in addition to React's features.

6-MARK

Q10 6 marks

Explain how HTML, CSS, and JavaScript each contribute to a library page that lists books, is styled, and filters results as the user types.

ANSWER

HTML defines the structure and content. CSS specifies appearance such as colors, layout, fonts, and sizes. JavaScript adds behavior such as filtering results while the user types and updating part of the screen.

THINK AS AN ENGINEER

Q11.1

A student club wants a page that lists events, uses the club's colors, and has a button showing how many people have signed up. Assign each feature to HTML, CSS, or JavaScript.

ANSWER

Event list → HTML. Colors and layout → CSS. Live sign-up counter → JavaScript.

THINK AS AN ENGINEER

Q11.2

Name one responsive-design change and one semantic-HTML choice that would help a phone user and a screen reader.

ANSWER

Use a responsive layout for the phone. Use meaningful elements such as <header>, <nav>, or <main> for the screen reader.

THINK AS AN ENGINEER

Q11.3

Should the club separate HTML, CSS, and JavaScript? Give two reasons.

ANSWER

Yes. Each technology has a clear role, and separation makes the page easier to understand, maintain, and develop.

NEW CONTEXT

Q12

A news site must look good on phones and be found easily by search engines. Give one responsive-design choice and one semantic-HTML choice, and state what each improves.

ANSWER

Use a responsive layout that re-flows content for small screens, improving usability. Use semantic elements such as `<header>`, `<nav>`, and `<main>`, improving accessibility and helping search engines understand the content.

REFLECT & CHALLENGE

Q13.1

Reflect: Of structure, style, and behavior, which do you notice most when a web page is badly made, and why?

ANSWER

Any can be noticeable: poor structure can confuse users, poor style can reduce readability, and poor behavior can make controls fail to respond.

REFLECT & CHALLENGE

Q13.2

Challenge: Name one website you use and describe one thing HTML, CSS, and JavaScript each do on it.

ANSWER

Example: HTML provides headings and links; CSS controls layout and appearance; JavaScript responds to clicks, typing, or other interactions.

REVIEW • CHAPTER 3 QUESTION BANK

Review**Chapter 3 — Mixed Revision Test**

Try these questions without looking at the answers.

FILL IN THE BLANK

QA1

Frontend + backend + _____ make the three-tier architecture.

ANSWER

database

FILL IN THE BLANK

QA2

The side that requests data or processing is the _____.

ANSWER

client

FILL IN THE BLANK

QA3

The communication protocol is _____.

ANSWER

HTTP

FILL IN THE BLANK

QA4

HTTP with SSL/TLS encryption is _____.

ANSWER

HTTPS

FILL IN THE BLANK

QA5

The method used to retrieve data is _____.

ANSWER

GET

FILL IN THE BLANK

QA6

The method used to send data is _____.

ANSWER

POST

FILL IN THE BLANK

QA7

Status code _____ means success.

ANSWER

200

FILL IN THE BLANK

QA8

Status code _____ means not found.

ANSWER

404

FILL IN THE BLANK

QA9

Status code _____ indicates a server-side error.

ANSWER

500

FILL IN THE BLANK

QA10

_____ defines structure.

ANSWER

HTML

FILL IN THE BLANK

QA11

_____ specifies appearance.

ANSWER

CSS

FILL IN THE BLANK

QA12

_____ adds behavior.

ANSWER

JavaScript

FILL IN THE BLANK

QA13

Layout adjustment for different screen sizes is _____.

ANSWER

responsive design

TRUE / FALSE

QB1

The database is the part users directly see. (True or False?)

ANSWER

False

TRUE / FALSE

QB2

A browser can act as a client. (True or False?)

ANSWER

True

TRUE / FALSE

QB3

HTTPS adds SSL/TLS encryption to HTTP. (True or False?)

ANSWER

True

TRUE / FALSE

QB4

GET is the appropriate method for submitting a contact form. (True or False?)

ANSWER

False — POST is appropriate.

TRUE / FALSE

QB5

404 means the requested page was not found. (True or False?)

ANSWER

True

TRUE / FALSE

QB6

JSON is easy for humans to read and computers to process. (True or False?)

ANSWER

True

TRUE / FALSE

QB7

CSS is mainly responsible for behavior. (True or False?)

ANSWER

False — JavaScript handles behavior.

TRUE / FALSE

QB8

Responsive design supports different screen sizes. (True or False?)

ANSWER

True

TRUE / FALSE

QB9

Semantic HTML can improve accessibility and SEO. (True or False?)

ANSWER

True

TRUE / FALSE

QB10

Frameworks require developers to write every common function from scratch. (True or False?)

ANSWER

False

MATCHING

QC1

Match each item with A Frontend, B Backend, or C Database.

- a. Product page
- b. Order processing
- c. Stored customer information
- d. Cart screen
- e. Stock checking
- f. Order history

ANSWER

- a → A (Frontend)
- b → B (Backend)
- c → C (Database)
- d → A (Frontend)
- e → B (Backend)
- f → C (Database)

MATCHING

QC2

Match each item with A HTML, B CSS, or C JavaScript.

- a. Headings and paragraphs
- b. Font size and colors
- c. Button changes screen
- d. Images and links
- e. Page layout
- f. Form validation

ANSWER

- a → A (HTML)
- b → B (CSS)
- c → C (JavaScript)
- d → A (HTML)
- e → B (CSS)
- f → C (JavaScript)

MULTIPLE CHOICE

QD1

Which part stores and manages data?

- A. Frontend
- B. Backend
- C. Database
- D. Browser

ANSWER

C

MULTIPLE CHOICE

QD2

Which HTTP method retrieves data?

- A. GET
- B. POST
- C. 404
- D. HTTPS

ANSWER

A

MULTIPLE CHOICE

QD3

Which status code indicates success?

- A. 200
- B. 404
- C. 500
- D. 301

ANSWER

A

MULTIPLE CHOICE

QD4

Which status code indicates a missing page?

- A. 200
- B. 404
- C. 500
- D. 100

ANSWER

B

MULTIPLE CHOICE

QD5

Which technology defines structure?

- A. CSS
- B. HTML
- C. JavaScript
- D. JSON

ANSWER

B

MULTIPLE CHOICE

QD6

Which technology controls appearance?

- A. HTML
- B. CSS
- C. JavaScript
- D. HTTP

ANSWER

B

MULTIPLE CHOICE

QD7

Which technology adds behavior?

- A. HTML
- B. CSS
- C. JavaScript
- D. Database

ANSWER

C

MULTIPLE CHOICE

QD8

Which is a benefit of semantic HTML?

- A. Necessarily faster display
- B. Better accessibility and SEO
- C. Replaces JavaScript
- D. Encrypts HTTP

ANSWER

B

MULTIPLE CHOICE

QD9

Which concept adapts layout to screen size?

- A. Responsive design
- B. API
- C. POST
- D. Database

ANSWER

A

MULTIPLE CHOICE

QD10

Which framework was developed by Meta?

- A. Vue
- B. React
- C. Next.js
- D. HTML

ANSWER

B

6-MARK

QE1 6 marks

Explain how a product search works through the three-tier architecture.

ANSWER

Frontend collects the keyword; backend receives and processes the request; database is searched; matching data returns to the backend; backend organizes the result; frontend displays it.

6-MARK

QE2 6 marks

Explain secure client-server communication when a user opens a web page.

ANSWER

The browser is the client and sends a request. HTTPS can carry the request securely because it adds SSL/TLS encryption to HTTP. The server processes the request and returns a response, which the browser displays.

6-MARK

QE3 6 marks

Explain HTML, CSS, and JavaScript on an interactive web page.

ANSWER

HTML defines structure/content, CSS controls appearance, and JavaScript adds behavior and interactivity.

THINK AS AN ENGINEER

QF1

A school wants a web application where students view events, submit registration forms, and receive confirmation. Which tier displays the event list and registration form?

ANSWER

Frontend.

THINK AS AN ENGINEER

QF2

Which tier processes the registration request?

ANSWER

Backend.

THINK AS AN ENGINEER

QF3

Which tier stores registration information?

ANSWER

Database.

THINK AS AN ENGINEER

QF4

Which HTTP method is appropriate for submitting the form?

ANSWER

POST.

THINK AS AN ENGINEER

QF5

Why use HTTPS?

ANSWER

It adds SSL/TLS encryption and helps protect communication content.

THINK AS AN ENGINEER

QF6

Which status code is appropriate if the requested page does not exist?

ANSWER

404.

NEW CONTEXT

QG1

A small online library wants a page that works well on smartphones, uses meaningful HTML elements, has an interactive search box, and receives book data from a server. Identify: (1) screen-size concept, (2) meaningful-HTML concept, (3) interactive technology, (4) common data format.

ANSWER

- (1) Responsive design.
- (2) Semantic HTML.
- (3) JavaScript.
- (4) JSON.

REFLECT & CHALLENGE

QH1

A page reports either that a resource cannot be found or that a server failed. Which two status codes would you investigate?

ANSWER

404 for not found; 500 for a server-side error.

REFLECT & CHALLENGE

QH2

Challenge: Describe one complete journey from a user's action to the final result using at least five terms from: frontend, backend, database, client, server, HTTP/HTTPS, GET/POST, status code, API, JSON.

ANSWER

Example: The client uses the frontend to enter a search. The frontend sends a GET request over HTTPS to the server. The backend processes it and uses the database. The server returns a response, possibly through an API using JSON, and the frontend displays the result.

REFERENCE • CHAPTER 3 QUESTION BANK

Teacher Answer Guide — Chapter 3

Memory Map

3-1

Frontend = shows/collects; Backend = decides/processes; Database = stores/retrieves.

3-2

Client requests; Server responds; HTTP communicates; HTTPS adds SSL/TLS encryption.

Methods — GET = retrieve; POST = send.

Status — 200 = success; 404 = not found; 500 = server-side error.

Data — API = software-to-software data exchange; JSON = readable data format.

3-3

HTML = structure; CSS = appearance; JavaScript = behavior/interactivity.

Modern frontend — Semantic HTML → accessibility + SEO; Responsive design → adapts to screen size;

Mobile-first → prioritizes small screens.

Frameworks — React = Meta; Vue = simple/incremental; Next.js = React features + mechanisms to speed page display.

Source basis: Chapter 3 of the provided Egyptian Baccalaureate textbook, lessons 3-1, 3-2, and 3-3, including the chapter's exercise and exam-style question formats.